$$\chi_{eff} = \frac{(1+q)}{M^2}\left[\vec{S_1}\cdot\hat{L} + \frac{1}{q}\vec{S_2}\cdot\hat{L}\right]$$

$$M\frac{d\chi_{eff}}{dt} = \frac{(1+q)}{M}\left[\underset{①}{\frac{d\vec{S_1}}{dt}}\cdot\hat{L} + \vec{S_1}\cdot\underset{②}{\frac{d\hat{L}}{dt}} + \frac{1}{q}\underset{③}{\frac{d\vec{S_2}}{dt}}\cdot\hat{L} + \frac{1}{q}\vec{S_2}\cdot\underset{④}{\frac{d\hat{L}}{dt}}\right]$$

① $$\frac{d\vec{S_1}}{dt} = \frac{1}{2r^3}\left\{\left[4+3q - \frac{3(1+q)^2}{\sqrt{r/M}}\left(K_1\vec{S_1} + \frac{\vec{S_2}}{q}\right)\cdot\frac{\hat{L}}{M^2}\right]\vec{L} + \vec{S_2}\right\}\times$$

$$\frac{1}{r^3}\left\{\left[4+3q - \frac{3(1+q)^2}{M^2\sqrt{r/M}}\left(K_2\vec{S_1}\cdot\hat{L} + \frac{\vec{S_2}}{q}\cdot\hat{L}\right)\right]\vec{L} + \vec{S_2}\right\}\times\vec{S_1}\cdot\hat{L}$$

$$\frac{1}{r^3}\left\{[\,''\quad''\,](\hat{L}\times\vec{S_1})\cdot\hat{L} + (\vec{S_2}\times\vec{S_1})\cdot\hat{L}\right. =$$

$$\frac{1}{r^3}\left\{[\,''\quad''\,](\hat{L}\times\hat{L})\cdot\vec{S_1} + (\vec{S_2}\times\vec{S_1})\cdot\hat{L}\right. =$$

(triple product + self cross product definition)

$$\frac{1}{r^3}\left\{(\vec{S_2}\times\vec{S_1})\cdot\hat{L}\right\}$$

② $$\frac{d\vec{L}}{dt} = \frac{1}{2r^3}\left\{[\,''\ ''\,](\vec{S_1}\times\vec{L}) + [\quad](\vec{S_2}\times\vec{L})\right\}$$

$$\frac{1}{2r^3}\left\{[\,''\ ''\,](\vec{S_1}\times\vec{L})\cdot\vec{S_1} + [\quad](\vec{S_2}\times\vec{L})\cdot\vec{S_1}\right\}$$

$$\frac{1}{2r^3}\left\{[\,''\ ''\,](S_1\times\vec{S_1})\cdot\vec{L} + [\quad](\vec{S_1}\times\vec{S_2})\cdot\vec{L}\right\}$$

$$\frac{1}{2r^3}\left\{[\quad](\vec{S_1}\times\vec{S_2})\cdot\hat{L}\right\}$$

③ $$\frac{d\vec{S_2}}{dt} = \frac{1}{2r^3}\left\{[\quad]\vec{L}\times\vec{S_2} + \vec{S_1}\times\vec{S_2}\right\}$$

$$\frac{1}{2r^3}\left\{[\quad](\vec{L}\times\vec{S_2})\cdot\hat{L} + (\vec{S_1}\times\vec{S_2})\cdot\hat{L}\right\} =$$

$$\frac{1}{2r^3}\left\{[\quad](\hat{L}\times\hat{L})\cdot\vec{S_2} + (\vec{S_1}\times\vec{S_2})\cdot\hat{L}\right\} =$$

$$\frac{1}{2r^3}\left\{(\vec{S_1}\times\vec{S_2})\cdot\hat{L}\right\}$$

④ $$\frac{1}{2r^3}\left\{[\,''\quad''\,](\vec{S_1}\times\vec{L})\cdot\vec{S_2} + [\quad](\vec{S_2}\times\vec{L})\cdot\vec{S_2}\right\}$$

$$\frac{1}{2r^3}\left\{[\,''\quad''\,](\vec{S_2}\times\vec{S_1})\cdot\hat{L} + [\quad](\vec{S_2}\times\vec{S_2})\cdot\hat{L}\right\}$$

$$\frac{1}{2r^3}\left\{[\,''\quad''\,](\vec{S_2}\times\vec{S_1})\cdot\hat{L}\right\}$$

All togheter

$$\frac{d\chi_{eff}}{dt} = (1+q)\left\{\frac{1}{2r^3}(\vec{S_2}\times\vec{S_1})\cdot\hat{L} + \frac{1}{2r^3}[\quad](\vec{S_1}\times\vec{S_2})\cdot\hat{L} + \frac{1}{q}\frac{1}{2r^3}(\vec{S_1}\times\vec{S_2})\cdot\hat{L} + \frac{1}{q}\frac{1}{2r^3}\left\{[\,''\quad''\,](\vec{S_2}\times\vec{S_1})\cdot\hat{L}\right\}\right.$$

$$= \frac{(1+q)}{2r^3}\left\{(\vec{S_2}\times\vec{S_1})\cdot\hat{L} + [\quad](\vec{S_1}\times\vec{S_2})\cdot\hat{L} + \frac{(\vec{S_1}\times\vec{S_2})\cdot\hat{L}}{q} + \frac{1}{q}[\,''\quad''\,](\vec{S_2}\times\vec{S_1})\cdot\hat{L}\right\}$$

$$(\vec{S_2}\times\vec{S_1})\cdot\hat{L} = -(\vec{S_1}\times\vec{S_2})\cdot\hat{L}$$

$$= \frac{(1+q)}{M\,2r^3}\left\{-(\vec{S_1}\times\vec{S_2})\cdot\hat{L} + [\quad](\vec{S_1}\times\vec{S_2})\cdot\hat{L} + \frac{1}{q}(\vec{S_1}\times\vec{S_2})\cdot\hat{L} - \frac{1}{q}[\,''\quad''\,](\vec{S_1}\times\vec{S_2})\cdot\hat{L}\right\}$$

$$= \frac{(1+q)}{M\,2r^3}\left\{-1 + [\quad] + \frac{1}{q} - \frac{1}{q}[\,''\quad''\,]\right\}(\vec{S_1}\times\vec{S_2})\cdot\hat{L}$$

$$[\quad] = 4 + \frac{3}{q} - \frac{3(1+q)^2}{q\sqrt{r/M}}\left(\frac{\vec{S_1}\cdot\hat{L}}{M^2} + \frac{K_2\vec{S_2}\cdot\hat{L}}{q\,M^2}\right)$$

$$[\,''\quad''\,] = 4 + 3q - \frac{3(1+q)^2}{M^2\sqrt{r/M}}\left(\vec{S_1}\cdot\hat{L}\,K_1 + \frac{\vec{S_2}\cdot\hat{L}}{q}\right)$$

$$\chi_{eff} = \frac{(1+q)}{M^2}\left[(\vec{S_1}\cdot\hat{L}) + \frac{1}{q}(\vec{S_2}\cdot\hat{L})\right]$$

$$\delta\chi = \frac{(1+q)}{M^2}\left[(\vec{S_1}\cdot\hat{L}) - \frac{1}{q}(\vec{S_2}\cdot\hat{L})\right]$$

$$\Downarrow$$

$$\chi_{eff} + \delta\chi = \frac{(1+q)}{M^2}\left[2(\vec{S_1}\cdot\hat{L})\right] \qquad (\vec{S_1}\cdot\hat{L}) = \frac{(\chi_{eff}+\delta\chi)M^2}{2(1+q)}$$

$$\chi_{eff} - \delta\chi = \frac{(1+q)}{M^2 q}\left[2(\vec{S_2}\cdot\hat{L})\right] \qquad \frac{(\vec{S_2}\cdot\hat{L})}{q} = \frac{(\chi_{eff}-\delta\chi)M^2}{2(1+q)}$$

$$\rightarrow 4 + 3q - \frac{3}{M^2}\frac{(1+q)^2 M^2}{\sqrt{r/M}}\left[\frac{(\chi_{eff}+\delta\chi)}{2(1+q)}K_1 + \frac{(\chi_{eff}-\delta\chi)}{2(1+q)}\right] =$$

$$4 + 3q - \frac{3}{2}\frac{(1+q)}{\sqrt{r/M}}\left[\chi_{eff}(K_1+1) + \delta\chi(K_1-1)\right]$$

$$\rightarrow 4 + \frac{3}{q} - \frac{3(1+q)^2 M^2}{M^2 q\sqrt{r/M}}\left(\frac{(\chi_{eff}+\delta\chi)}{2(1+q)} + K_2\frac{(\chi_{eff}-\delta\chi)}{2(1+q)}\right) =$$

$$4 + \frac{3}{q} - \frac{3}{2}\frac{(1+q)}{q\sqrt{r/M}}\left(\chi_{eff}(1+K_2) + \delta\chi(1-K_2)\right)$$

$$\circledast\ \frac{(1+q)}{M\,2r^3}\left\{-1 + 4 + \frac{3}{q} - \frac{3}{2}\frac{(1+q)}{q\sqrt{r/M}}\left[\chi_{eff}(1+K_2) + \delta\chi(1-K_2)\right] + \frac{1}{q} - \frac{1}{q}\left[4 + 3q - \frac{3}{2}\frac{(1+q)}{\sqrt{r/M}}\left(\chi_{eff}(K_1+1) + \delta\chi(K_1-1)\right)\right.\right.$$

$$= \frac{1}{2r^3}\left\{+\frac{3}{2}\frac{(1+q)}{\sqrt{r/M}}\frac{1}{q}\left[-\chi_{eff}(1+K_2) - \delta\chi(1-K_2) + \chi_{eff}(K_1+1) + \delta\chi(K_1-1)\right]\right\} =$$

$$= \frac{3}{4r^3}\frac{(1+q)}{qM\sqrt{r/M}}\left[-\chi_{eff} - \chi_{eff}K_2 - \delta\chi + \delta\chi K_2 + \chi_{eff}K_1 + \chi_{eff} + \delta\chi K_1 - \delta\chi\right]$$

$$= \frac{3}{4r^3}\frac{(1+q)}{qM}\frac{1}{\sqrt{r/M}}\left[+\chi_{eff}(K_1-K_2) + \delta\chi(K_2+K_1-2)\right]$$

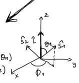
$$\frac{d\chi_{eff}}{dt} = \frac{3}{4r^3}\frac{(1+q)^2}{qM}\frac{1}{\sqrt{r/M}}\left[\chi_{eff}(K_1-K_2) + \delta\chi(K_2+K_1-2)\right](\vec{S_1}\times\vec{S_2})\cdot\hat{L}$$

$$\vec{S_1} = \frac{\chi_1}{(1+q)^2}M^2\hat{S_1} \qquad \vec{S_2} = \frac{\chi_2 q^2}{(1+q)^2}M^2\hat{S_2}$$

$$\frac{d\chi_{eff}}{dt} = \frac{3}{4r^3}\frac{(1+q)^2}{qM}\frac{1}{\sqrt{r/M}}\frac{\chi_1\chi_2 M^4 q^2}{(1+q)^4}[\sim\sim](\hat{S_1}\times\hat{S_2})\cdot\hat{L}$$

$$= \frac{3}{4}\frac{\chi_1\chi_2 q M^3}{\sqrt{r/M}\,r^3}\frac{1}{(1+q)^2}[\quad]\ \sin\theta_1\sin\theta_2\sin\Delta\phi$$

$$(\hat{S_1}\times\hat{S_2})\cdot\hat{L}$$



$$\hat{L} = (0,0,1)$$

$$\hat{S_1} = (\sin\theta_1\cos\phi_1,\ \sin\theta_1\sin\phi_1,\ \cos\theta_1)$$

$$\hat{S_2} = (\sin\theta_2\cos\phi_2,\ \sin\theta_2\sin\phi_2,\ \cos\theta_2)$$

$$\hat{L}\cdot(\hat{S_1}\times\hat{S_2}) = \hat{L}_x(S_{1y}S_{2z} - S_{1z}S_{2y}) + \hat{L}_y(S_{1z}S_{2x} - S_{1x}S_{2z}) + \hat{L}_z(S_{1x}S_{2y} - S_{1y}S_{2x}) =$$

$$= \hat{L}_z(S_{1x}S_{2y} - S_{1y}S_{2x}) =$$

$$= (\sin\theta_1\cos\phi_1\sin\theta_2\sin\phi_2 - \sin\theta_1\sin\phi_1\sin\theta_2\cos\phi_2)$$

$$= \sin\theta_1\sin\theta_2(\cos\phi_1\sin\phi_2 - \sin\phi_1\cos\phi_2)$$

$$\sin(A-B) = \sin A\cos B - \cos A\sin B \qquad \sin(\phi_2-\phi_1)$$

$$= \sin\theta_1\sin\theta_2\sin\Delta\phi \qquad \Delta\phi = \phi_2 - \phi_1$$

$$\frac{\chi_1 M^2\cos\theta_1}{(1+q)^2} \qquad \frac{q\chi_2 M^2}{(1+q)^2}$$

$$\delta\chi = \frac{(1+q)}{M^2}\left(\vec{S_1}\cdot\hat{L} - \frac{\vec{S_2}\cdot\hat{L}}{q}\right)$$

$$\frac{d\delta\chi}{dt} = \frac{(1+q)}{M^2}\left(\underset{①}{\frac{d\vec{S_1}}{dt}}\cdot\hat{L} + \underset{②}{\frac{d\hat{L}}{dt}}\cdot\vec{S_1} - \underset{③}{\frac{d\vec{S_2}}{dt}}\cdot\frac{\hat{L}}{q} - \underset{④}{\frac{d\hat{L}}{dt}}\cdot\frac{\vec{S_2}}{q}\right.$$

We have already calculated all there parts

$$\frac{d\delta\chi}{dt} = \frac{(1+q)}{M^2}\left[\frac{1}{2r^3}\left\{(\vec{S_2}\times\vec{S_1})\cdot\hat{L}\right\} + \frac{1}{2r^3}\left\{[\quad](\vec{S_1}\times\vec{S_2})\cdot\hat{L}\right\} + -\frac{1}{2r^3}\frac{1}{q}\left\{(\vec{S_1}\times\vec{S_2})\cdot\hat{L}\right\} + -\frac{1}{2r^3 q}\left\{[\,''\quad''\,](\vec{S_2}\times\vec{S_1})\cdot\hat{L}\right\}\right]$$

$$= \frac{(1+q)}{2r^3M^2}\left\{-(\vec{S_1}\times\vec{S_2})\cdot\hat{L} + [\quad](\vec{S_1}\times\vec{S_2})\cdot\hat{L} - \frac{(\vec{S_1}\times\vec{S_2})\cdot\hat{L}}{q} + (\vec{S_1}\times\vec{S_2})\cdot\hat{L}[\,''\quad''\,]\frac{1}{q}\right\}$$

$$= \frac{(1+q)}{2r^3M^2}\left\{-1 - \frac{1}{q} + [\quad] + \frac{1}{q}[\,''\quad''\,]\right\}(\vec{S_1}\times\vec{S_2})\cdot\hat{L}$$

$$[\,''\quad''\,] = 4 + 3q - \frac{3}{2}\frac{(1+q)}{\sqrt{r/M}}\left[\chi_{eff}(K_1+1) + \delta\chi(K_1-1)\right]$$

$$[\quad] = 4 + \frac{3}{q} - \frac{3}{2}\frac{(1+q)}{q\sqrt{r/M}}\left(\chi_{eff}(1+K_2) + \delta\chi(1-K_2)\right)$$

$$\frac{(1+q)}{2r^3}\left\{-1 - \frac{1}{q} + 4 + \frac{3}{q} - \frac{3}{2}\frac{(1+q)}{q\sqrt{r/M}}(\quad) + \frac{4}{q} + 3 - \frac{3}{2}\frac{(1+q)}{qM^2\sqrt{r/M}}(\quad)\right\}$$

$$\frac{(1+q)}{2r^3}\left\{-1 + 4 + 3 - \frac{1}{q} + \frac{3}{q} + \frac{4}{q}\right.$$

$$6 + \frac{6}{q} - \frac{3}{2}\frac{(1+q)}{q\sqrt{r/M}}\left[\chi_{eff}K_1 + \chi_{eff} + \delta\chi K_1 - \delta\chi + \chi_{eff} + \chi_{eff}K_2 + \delta\chi - K_2\delta\chi\right.$$

$$= \frac{(1+q)}{2r^3}\left\{6 + \frac{6}{q}\quad\frac{3(1+q)}{q\sqrt{r/M}}\left[2\chi_{eff} + \chi_{eff}K_1 + \chi_{eff}K_2 + \delta\chi K_1 - \delta\chi\right.\right.$$

$$= \frac{(1+q)}{2r^3}\left\{6\quad - \frac{3}{2}\frac{(1+q)}{q\sqrt{r/M}}\left[\chi_{eff}(2+K_1+K_2) + \delta\chi(K_1-K_2)\right]\right.$$

$$= \frac{(1+q)}{2r^3}\left\{\frac{6(1+q)}{q} - \frac{3}{2}\frac{(1+q)}{q\,r^{1/2}/M^{1/2}}[\quad]\right\} =$$

$$= \frac{(1+q)}{2r^3}\frac{1}{M}\frac{6(1+q)}{q}\left[1 - \frac{1}{4\sqrt{r/M}}\left[\chi_{eff}(2+K_1+K_2) + \delta\chi(K_1-K_2)\right]\right]$$

$$= \frac{3(1+q)^2}{Mq\,r^3}\frac{\chi_1\chi_2 M^4 q^2}{(1+q)^4}\left[1 - \frac{1}{4\sqrt{r/M}}(\quad)\right]$$

$$= \frac{3\chi_1\chi_2 M^3}{(1+q)^2 r^3}q\left[1 - \frac{1}{4\sqrt{r/M}}\left(\chi_{eff}(2+K_1+K_1) + \delta(K_1\right.\right]$$

$$\frac{3q\chi_1\chi_2}{(1+q)^2}\left(\frac{r}{M}\right)^{-3}\left[1 - \frac{\chi_{eff}}{\sqrt{r/M}}\right]\sin\theta_1\sin\theta_2\sin\Delta\phi$$

$$K_1 = K_2 = 1$$

$$M\frac{d\delta\chi}{dt} = \frac{3q\chi_1\chi_2}{(1+q)^2}\left(\frac{M}{r}\right)^3\left\{1 - \frac{[\chi_{eff}(2+K_1+K_2) + \delta\chi(K_1-K_2)]}{4\sqrt{r/M}}\right\}\odot$$

$$M\frac{d\chi_{eff}}{dt} = \frac{3}{4}\frac{q\chi_1\chi_2}{r^3\sqrt{r/M}}\frac{M^3}{(1+q)^2}\left[\chi_{eff}(K_1-K_2) + \delta\chi(K_2+K_1-2)\right]\odot$$

$$\odot = \sin\theta_1\sin\theta_2\sin\Delta\phi$$